PHY 211: ELEMENTARY MODERN PHYSICS
WAVE FUNCTION

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INTRODUCTION

- The postulates are the **axioms** on which quantum theory is based.
- They are basic premises of the (mathematical) theory, which are assumed valid without mathematical proof because adopting them in the calculations based on the theory is the only means by which the theoretical results can make physical sense (i.e. agree with experimental results).

Postulates of Quantum Mechanics

Postulate I: The state of a particle is specified by wavefunction $\Psi(\vec{r}, t)$, a (complex) function of position vector, $\vec{r}$, and time, t.

Postulate II: For every classical observable, there corresponds a linear, hermitian, quantum mechanical operator.

Postulate III: Any measurement "A" will only yield eigenvalues of $\hat{A}$

$$\hat{A}\psi_n = a_n\psi_n$$

Postulate IV: The wave function evolves in time according to the time-dependent Schrödinger Equation.

$$\hat{H}\Psi(r, t) = i\hbar\frac{\partial\Psi(r, t)}{\partial t}$$

Postulate V: If a system is described by a wave function ψ, then the average value is given as

$$\langle A\rangle = \frac{\int_{-\infty}^{+\infty}\psi^*\hat{A}\psi d\tau}{\int_{-\infty}^{+\infty}\psi^*\psi d\tau}$$

- In quantum physics, the state of a particle is described by a wave function, $\Psi(r,t)$. The wave function describes the de Broglie wave of a particle, giving its amplitude as a function of position and time.
- All matter have wave properties with the particle. The wave quantities are related by de Broglie relations:

$$p = \frac{h}{\lambda}$$
$$E = hf$$

- The function describing how light waves propagate is the electromagnetic wave function $E(x,t) = E_{max} sin(ax - bt)$ with intensity $\propto E_{max}^2 \propto E_{avg}$.

Particles also have a wave function $\Psi(x)$. In this case, $|\Psi(x)|^2$ is the probability density which indicates how likely the particle is to be found at x (or what fraction of the particles will be found at x).

NOTE: The wave function gives a particle's amplitude, not intensity; if you want to find the intensity of the wave function, you have to square it: $|\Psi(r,t)|^2$. The intensity of a wave is what is equal to the probability that the particle will be at that position at that time.

$|\Psi(r,t)|^2 d^3r$ is the probability that the particle will be found in the volume element d^3r, located at position r at time t.

Besides the position-space wave function $\Psi(r,t)$, there is also a momentum-space version of the wave function: $\Phi(p,t)$.

- Max Born (Nobel Prize, 1954), one of the founders of the quantum theory, first proposed this probability interpretation of the wave function in 1926. The link between a particle's behaviour and its associated wave function is given by Max Born's *statistical interpretation* of the wave function.

Max Born's postulate

If a measurement is made at time t to locate the particle associated with the wave function $\Psi(x,t)$, the the probability $P(x,t)dx$ that the particle will be found between x and $x+dx$ $[\Psi(x,t)]$ is equal to $\Psi^*(x,t)\Psi(x,t)$

- The use of wave functions $\Psi(x, y, z, t)$ rather than the classical positions $x(t)$, $y(t)$, $z(t)$ represents a clean break between classical and modern physics.
- Erwin Schrödinger and Werner Heisenberg worked out independent and separate mathematical models for the quantum theory in 1926.
- There are various formulations of the interpretation of quantum theory, but it is generally based on the following:
  1. The uncertainty principle of Heisenberg
  2. The complementarity principle of Bohr
  3. The statistical interpretation of Born, based on probabilities determined by the wave function

  Together these three concepts form a logical interpretation of the physical meaning of quantum theory.

WAVE FUNCTION AND ITS PROPERTIES

A wave function in quantum mechanics is a description of the quantum state of a system.The wave function is a complex-valued probability amplitude, and the probabilities for the possible results of measurements made on the system can be derived from it.

- Wave functions are complex valued and are **not** directly observable. Probability density is observable:

$$|\psi(x)|^2 = \psi^*(x)\psi(x) = \psi^2_{real} + \psi^2_{imag}$$

- The probability of finding the particle at some infinitesimal distance δx around x is

$$|\psi(x)|^2 \delta x$$

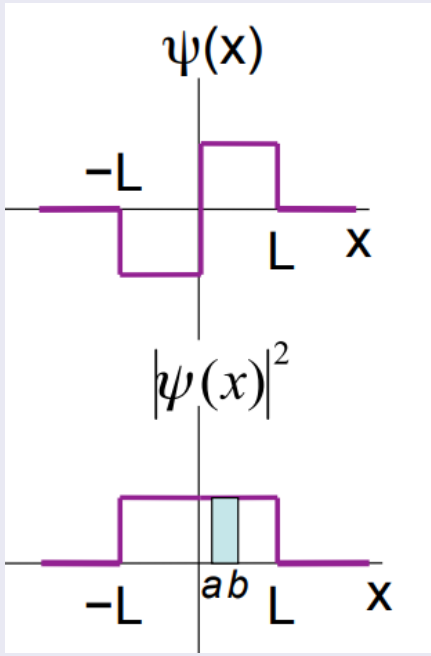
- The particle must be somewhere with a probability of 100%.

$$\int_{-\infty}^{+\infty} |\psi(x)|^2 dx = 1$$

This is the normalization condition.

NOTE: $\Psi(x,t)$ is the full wave function while $\psi(x)$ is just the spatial part of the wave function.

In 1D, $|\psi(x)|^2$ has dimension of 1/Length.



- The localization of a wave can be accomplished by using a wave packet. We used a function $\Psi(x,t)$ to denote the superposition of many waves to describe the wave packet.

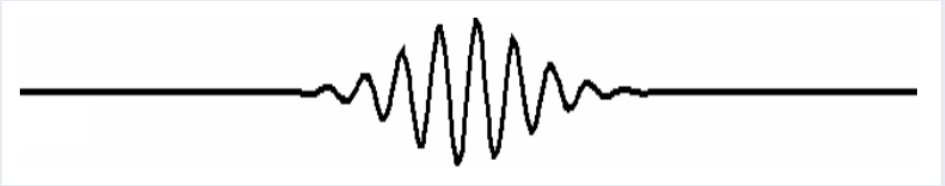
Plane Waves vs. Wave Packets

Plane Wave: This wave represents a single k and ω. Therefore energy, momentum, and wavelength are well defined. The amplitude is the same over all space and time, so *position* and *time* are **undefined**.



$$\Psi(x,t) = Ae^{i(kx-\omega t)}$$

Wave Packet: This wave is composed of many different k and ω waves. Thus, it is composed of many different energies, momenta, and wavelengths and so these quantities are not well defined. The amplitude is non-zero in a small region of space and time so the position and time is constrained to be in that region.



$$\Psi(x,t) = \sum_{n=0}^{\infty} A_n e^{i(k_n x-\omega_n t)}$$

1D Schrödinger Equation

The wave equation for electromagnetic waves is

$$\frac{\partial^2 E}{\partial x} = \frac{1}{c^2}\frac{\partial^2 E}{\partial t}$$

Only works for massless particles with $v = c$.

For massive particles, we need the time dependent Schrödinger equation (TDSE):

$$\frac{-\hbar^2}{2m}\frac{\partial^2 \Psi(x,t)}{\partial x^2} + V(x,t)\Psi(x,t) = i\hbar\frac{\partial \Psi(x,t)}{\partial t}$$

where $i = \sqrt{-1}$ is an imaginary number and we have used partial derivatives. Both the potential V and wave function Ψ may be functions of space and time, $V(x,t)$ and $\Psi(x,t)$.

3D Schrödinger Equation

The extension of time dependent Schrödinger equation to three dimensions is fairly straight:

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\Psi}{\partial x^2}+\frac{\partial^2\Psi}{\partial y^2}+\frac{\partial^2\Psi}{\partial z^2}\right)+V(x,y,z,t)\Psi(x,y,z,t)=i\hbar\frac{\partial\Psi(x,y,z,t)}{\partial t}$$

$$-\frac{\hbar^2}{2m}\nabla^2\Psi+V\Psi=i\hbar\frac{\partial\Psi}{\partial t}$$

where

$\nabla^2=\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}$ For Cartesian coordinate

** $i=\sqrt{-1}, i^2=-1, \frac{1}{i}=-i$

For time independent potentials, $V(x)$: $\Psi(x,t) = \psi(x)\phi(t)$

In this case, the time part of the wave function is: $\phi(t) = e^{-iEt/\hbar}$

The spatial part $\psi(x)$ can be found from the time independent Schrödinger equation (TISE):

$$\frac{-\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + V(x)\psi(x) = E\psi(x)$$

EIGENSTATE

- In general, the Schrodinger equation is

$$H\Psi = E\Psi$$

- H is the Hamiltonian operator, i.e., the energy operator.

$$E = \frac{H\Psi}{\Psi}$$

- Schrodinger equation is an eigenvalue equation

(Operator) $\times$ (Function) = (Constant) $\times$ (Same Function)
where the **Function** is called the **Eigenfunction**;
and the **Constant** is called **Eigenvalue**

Example on SWE

Show that $\Psi = Ae^{i(kx-\omega t)}$ satisfies the time-dependent Schrödinger wave equation.

Strategy: We again take the derivatives needed for the SWE Equation to see whether it is satisfied.

Solution

$$\Psi = Ae^{i(kx-\omega t)}$$

$$\frac{\partial \Psi}{\partial t} = -i\omega Ae^{i(kx-\omega t)}$$

$$= -i\omega\Psi$$

Also,

$$\Psi = Ae^{i(kx-\omega t)}$$

$$\frac{\partial \Psi}{\partial x} = ikAe^{i(kx-\omega t)}$$

$$= ik\Psi$$

And

$$\Psi = Ae^{i(kx-\omega t)}$$

$$\frac{\partial^2 \Psi}{\partial x^2} = \frac{\partial(ik\Psi)}{\partial x}$$

$$\frac{\partial^2 \Psi}{\partial x^2} = ik\frac{\partial(\Psi)}{\partial x}$$

$$= ik(ikAe^{i(kx-\omega t)})$$

$$= -k^2\Psi$$

Recall SWE

$$i\hbar\frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

Make all necesaary substitution, we

$$i\hbar(-i\omega\Psi) = -\frac{\hbar^2}{2m}(-k^2\Psi) + V\Psi$$

$$\hbar\omega\Psi = \frac{\hbar^2 k^2}{2m}\Psi + V\Psi$$

$$\hbar\omega\Psi - \frac{\hbar^2 k^2}{2m}\Psi - V\Psi = 0$$

where Energy is: $E = hf = \frac{h\omega}{2\pi} = \hbar\omega$

The wavenumber $k = \frac{2\pi}{\lambda} = \frac{2\pi}{h/P} = \frac{2\pi P}{h} = \frac{P}{\hbar}$

Momentum $P = \hbar k$ Therefore

$$E - \frac{P^2}{2m} - V = 0$$
$$E - K - V = 0$$
$$K + V = E$$

Thus, $\Psi = Ae^{i(kx-\omega t)}$ is a good solution and satisfies the Schrödinger equation.

Class Activity: Theoretical Calculation (Prediction) of the Result of a Measurement

Calculation of Probability of Finding a Particle in a Region

The state of a 1D single-particle system is described by

$$\Psi = a^{-\frac{1}{2}} e^{-\frac{|x|}{a}}$$

at $t = 0$, where $a = 1.0000nm$.
(a) Check that Ψ is normalised.
If system's state (position of particle) is measured at t = 0,
(b) Find the probability that the measured value will lie between $x = 1.5000nm$ and $x = 1.5001nm$;
(c) Find the probability that the measured value will lie between $x = 0.0000nm$ and $x = 2.000nm$.

Solution: Checking if $\Psi = a^{-\frac{1}{2}} e^{-\frac{|x|}{a}}$ is normalised.

$$\int_{-\infty}^{+\infty} \Psi^* \Psi dx = \int_{-\infty}^{+\infty} |\Psi(x)|^2 dx = 1$$

$$\begin{aligned}
\int_{-\infty}^{+\infty} \Psi^* \Psi dx &= \int_{-\infty}^{+\infty} a^{-\frac{1}{2}} e^{-\frac{|x|}{a}} \, . \, a^{-\frac{1}{2}} e^{-\frac{|x|}{a}} dx \\
&= a^{-1} \int_{-\infty}^{+\infty} e^{-2\frac{|x|}{a}} dx \\
&= a^{-1} \left[\int_{-\infty}^{0} e^{-2\frac{(-x)}{a}} dx + \int_{0}^{+\infty} e^{-2\frac{(+x)}{a}} dx \right] \\
&= a^{-1} \left[\frac{1}{\frac{2}{a}} e^{2\frac{(x)}{a}} \Big|_{-\infty}^{0} + \frac{1}{-\frac{2}{a}} e^{-2\frac{(x)}{a}} \Big|_{0}^{+\infty} \right] \\
&= a^{-1} \cdot \frac{a}{2} \left[e^{\frac{2x}{a}} \Big|_{-\infty}^{0} - e^{\frac{-2x}{a}} \Big|_{0}^{+\infty} \right]
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{2}\left[\left(e^{\frac{2(0)}{a}} - e^{\frac{2(-\infty)}{a}}\right) - \left(e^{\frac{-2(\infty)}{a}} - e^{\frac{-2(0)}{a}}\right)\right] \\
&= \frac{1}{2}\left[\left(e^{0} - e^{-\infty}\right) - \left(e^{-\infty} - e^{0}\right)\right] \\
&= \frac{1}{2}\left[(1-0) - (0-1)\right] \\
&= \frac{1}{2}.2 \\
&= 1
\end{aligned}
$$

$\therefore \int_{-\infty}^{+\infty} \Psi^* \Psi dx = 1$ i.e. Ψ is normalized

Assignment

Solve the part (b) and (c) of the given example